

# Polynomials (Homework)

Name  
Date  
Course

Simplify the following polynomials, write in descending order of powers, and state the degree.

①  $5x^6 + 2x^2 - 3x^6 + 6x - 7x^2 + 2$

②  $-5x - 2x^3 + 10 + 3x^2 - 2x - 9 + 8x^3 - 2x + 5x + 10x^2$

③  $2 + x^2 - 8x^2 + 3x^2 + 6x + 4 - 8x + x^2 - 5x^5$

④  $-x^3 + 2x - 4x + 8 - 3 + 2x - 1$

⑤  $100x - 7x^2 + 2 - 4 - 8 + x^2 + 4x - 8x$

Add or subtract the following polynomials.

⑥  $(2x^3 - 3x^2 + 7x + 5) + (5x^2 - 6x + 2)$

⑦  $(7x^3 + 3x^2 - 2x + 9) - (4x^3 - 5x^2 + 2x - 3)$

⑧  $(12x^3 - 6x^2 + 7x + 5) - (5x^4 - x^3 + 2x^2 - 2x - 1)$

⑨  $(x^3 - 3x^2 + 7x + 5) + (9x + 2)$

⑩  $(-x^3 + 3x) - (-8x^3 + 9x + 2)$

⑪  $(-6x^2 + 2x - 1) - (2x^4 + 3x - 8)$

⑫  $(4x - 6) + (-3x^2 + 9x - 1)$

⑬  $(7x^3 + 4x^2 - 2x + 5) - (3x^2 - 4x + 8)$

⑭  $(-8x^3 + x^2) + (-4x^3 - 8x^2 - 7x + 13)$

Multiply.

⑮  $4x^3(2x^2 - 3x + 9)$

⑯  $-2z^5(3z^4 - 8z^3 - 5z^2 + 7z - 2)$

⑰  $7y^5(2y^3 + 4y^2 - y + 1)$

⑱  $-6x^7(3x^2 - 2x + 10)$

⑲  $y^5(2y^3 + 3y^2 - 4y + 6)$

⑳  $-z^8(2z^3 + 6z^2 - z + 6)$

㉑  $3x(4x^2 - 2x + 10)$

㉒  $-10x^3(7x^3 - 7x^2 + 2)$

Divide.

㉓ 
$$\frac{20x^5 - 35x^4 + 10x^3 - 30x^2}{5x}$$

㉔ 
$$\frac{64x^3 - 40x^2}{8x^2}$$

㉕ 
$$\frac{100x^9 + 50x^5 - 80x^3 + 20x^2}{-10x}$$



$$\textcircled{26} \frac{15x^2 + 3x + 9}{3}$$

$$\textcircled{27} \frac{-56x^5 - 49x^4 - 63x^3}{-7x}$$

$$\textcircled{28} \frac{81x^4 - 45x^3 + 36x^2 - 72x}{9x}$$

$$\textcircled{29} \frac{7x^3 - 8x^2 + 3x}{x}$$

Answers

$$\textcircled{1} 2x^6 - 5x^2 + 6x + 2, D=6$$

$$\textcircled{2} 6x^3 + 13x^2 - 4x + 1, D=3$$

$$\textcircled{3} -5x^5 - 3x^2 - 2x + 6, D=5$$

$$\textcircled{4} -x^3 + 4, D=3$$

$$\textcircled{5} -6x^2 + 96x - 10, D=2$$

$$\textcircled{6} 2x^3 + 2x^2 + x + 7$$

$$\textcircled{7} 3x^3 + 8x^2 - 4x + 12$$

$$\textcircled{8} -5x^4 + 13x^3 - 8x^2 + 9x + 6$$

$$\textcircled{9} x^3 - 3x^2 + 16x + 7$$

$$\textcircled{26} 5x^2 + x + 3$$

$$\textcircled{10} 7x^3 - 6x - 2$$

$$\textcircled{27} 8x^4 + 7x^3 + 9x^2$$

$$\textcircled{11} -2x^4 - 6x^2 - x + 7$$

$$\textcircled{28} 9x^3 - 5x^2 + 4x - 8$$

$$\textcircled{12} -3x^2 + 13x - 7$$

$$\textcircled{29} 7x^2 - 8x + 3$$

$$\textcircled{13} 7x^3 + x^2 + 2x - 3$$

$$\textcircled{14} -12x^3 - 7x^2 - 7x + 13$$

$$\textcircled{15} 8x^5 - 12x^4 + 36x^3$$

$$\textcircled{16} -6z^9 + 16z^8 + 10z^7 - 14z^6 + 4z^5$$

$$\textcircled{17} 14y^8 + 28y^7 - 7y^6 + 7y^5$$

$$\textcircled{18} -18x^9 + 12x^8 - 60x^7$$

$$\textcircled{19} 2y^8 + 3y^7 - 4y^6 + 6y^5$$

$$\textcircled{20} -2z^{11} - 6z^{10} + z^9 - 6z^8$$

$$\textcircled{21} 12x^3 - 6x^2 + 30x$$

$$\textcircled{22} -70x^6 + 70x^5 - 20x^3$$

$$\textcircled{23} 4x^4 - 7x^3 + 2x^2 - 6x$$

$$\textcircled{24} 8x - 5$$

$$\textcircled{25} -10x^8 - 5x^4 + 8x^2 - 2x$$